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-----French-----

Exercice 3

- 1) Ecrire la fonction `int ** alloue_matrice_pascal (int n)` qui alloue la mémoire pour une matrice triangulaire de degré n lu au clavier.
- 2) Ecrire une fonction `int ** remplit_matrice_pascal (int n)`, qui stocke les coefficients binomiaux d'un polynôme de degré n (matrice de Pascal de degré n).
- 3) Ecrire une fonction `void affiche_matrice_pascal(int n)` qui affiche une matrice de pascal de degré n.
- 4) Ecrire un programme principal de test.

-----English-----

Exercise 3

- 1) Write the function `int ** alloue_matrice_pascal (int n)` which allocates the memory for a triangular matrix of degree n read from the keyboard.
- 2) Write a function `int ** fill_matrice_pascal (int n)` which stores the binomial coefficients of a polynomial of degree n (Pascal matrix of degree n).
- 3) Write a void function `displays_matrix_pascal(int n)` which displays a pascal matrix of degree n.
- 4) Write a main test program.

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#include <stdio.h>
#include <stdlib.h>

/* Functions in use */

int **allocate(int rows);
int **get_values(int **pascal, int rows);
void display(int **pascal, int rows);

int main()
{
    int **pascal, rows;
    printf("Rows = ");
    scanf("%d", &rows);
    pascal = allocate(rows);
    get_values(pascal, rows);
    display(pascal, rows);

    return 0;
}

int **allocate(int rows)
{
    int **pascal, i;
    pascal = (int **)malloc(rows * sizeof(int *));
    for (i = 0; i < rows; i++)
        pascal[i] = (int *)malloc(rows * sizeof(int));
    return pascal;
}

int **get_values(int **pascal, int rows)
{
    int i, j;
    for (i = 0; i < rows; i++)
    {
        for (j = 0; j <= i; j++)
            if (i == j || j == 0)
                (*(pascal + i) + j) = 1;
        else

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        *(*(pascal + i) + j) = *(*(pascal + i - 1) + (j - 1)) + *(*(pascal + i
- 1) + j);
    }
    return pascal;
}

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void display(int **pascal, int rows)
{
    int i, j, space;
    printf("Displaying:_____\\n");
    for (i = 0; i < rows; i++)
    {
        for (j = 0; j <= i; j++)
            printf("%d ", *(*(pascal + i) + j));
        printf("\\n");
    }
}

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